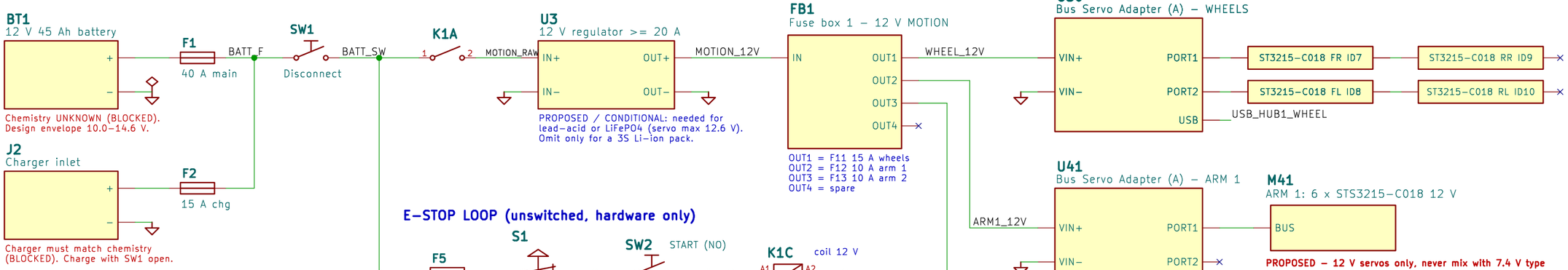
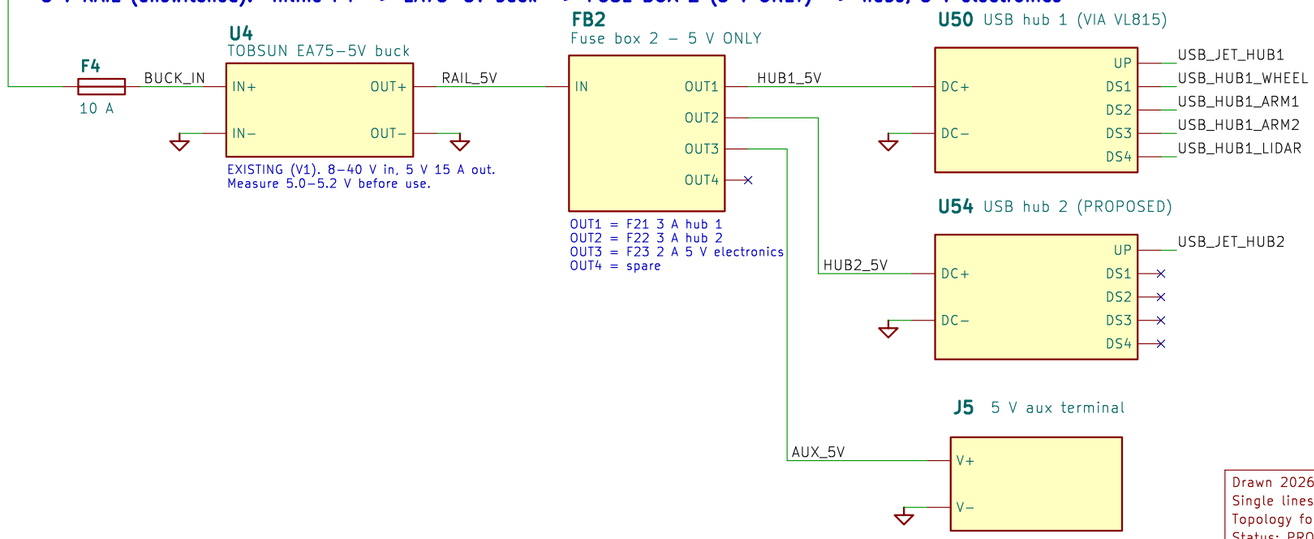
HAEBOT / KIRO V2-B 12 V 45 Ah EV BATTERY POWER SYSTEM

Electrical schematic, Rev A, 2026-11-13. STATUS: PROPOSED - battery chemistry, BMS limits and charger are BLOCKED items

MOTION POWER (cut by E-stop): battery -> F1 -> SW1 -> K1 -> 12 V regulator -> FUSE BOX 1 -> wheels, arm 1, arm 2



5 V RAIL (unswitched): inline F4 -> EA75-5V buck -> FUSE BOX 2 (5 V ONLY) -> hubs, 5 V electronics



SYMBOLS AND STATUS

Arrow-down symbol = POWER_GND:
own return wire to the star point
(negative bus bar) at battery -.
Never return motor current
through a USB cable.

EXISTING = on the V1 robot
PROPOSED = not bought / tested
BLOCKED = needs a hardware fact

E-STOP BEHAVIOUR

S1 pressed or wire broken:
K1 opens, wheels + both arms
lose power (arms drop).
Jetson, hubs, 5 V stay on.
Release S1: no restart until
START is pressed.

BEFORE BUILDING

Read battery + charger labels.
Fix F1, F2, K1, U3 after that.

On hub 1, port DS4:
USB_HUB1_LIDAR -> RPLIDAR C1 + CP2102N

DESIGN VALUES

Jetson 12 V: 25 / 36 / 45 W
Wheels: 1.2 / 6.0 / 10.8 A
Arm (12 V, each): 3 A design,
8 A peak (ESTIMATE)
5 V rail: about 10 W (assumed)
Battery: see docs/power_budget.md

WIRE (from V1 Rev F rules)

Battery to FB1: 8 AWG
Wheels, arms, buck: 14 AWG
Jetson: 18 AWG
Control loop: 20 AWG

WARNINGS

1. Never feed FB2 from 12 V.
2. Never connect 7.4 V servos
(limit 8.0 V) to FB1.
3. ST3215 max 12.6 V: battery
may reach 14.6 V -> U3.

Drawn 2026-11-13. KiCad ERC must be run in KiCad 8+ before release
Single lines to servos = 3-wire Molex 5264 bus cable (GND / VCC / SIG)
Topology follows V1 Rev F (two fuse boxes, EA75-5V buck); SMPS replaced by battery
Status: PROPOSED - battery chemistry, BMS and charger are not verified (BLOCKED items)
Genie EduTech

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Title: HAEBOT / KIRO V2-B - 12 V 45 Ah EV battery power system

Size: A3	Date: 2026-11-13	**Rev: A**
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